ENGINEERING WORKSHOP LAB

Semester Project Report

Automatic Water Dispenser

A Smart, Sensor-Based Arduino Project

Field	Details
Project Title	Automatic Water Dispenser
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1. Abstract

This report presents the design, development, and implementation of an Automatic Water Dispenser system built using an Arduino UNO microcontroller. The system enables users to dispense precise volumes of water 100 ml, 150 ml, or 200 ml through a simple button-controlled OLED menu interface. A submersible water pump is activated automatically upon detection of a glass via an Ultrasonic Sensor, and a Water Level Sensor continuously monitors the reservoir to alert users when water is running low. This project demonstrates the practical integration of multiple electronic components and embedded programming to solve a real-world problem in a smart and automated manner.

2. Introduction

In everyday life, dispensing an accurate quantity of liquid manually is often imprecise and inconvenient. Whether it is for medicinal purposes, cooking, or general hydration, the need for a reliable, automated dispensing mechanism is evident. The Automatic Water Dispenser project addresses this challenge by combining sensor-based detection, a microcontroller, and a pump mechanism to create a system that dispenses exact amounts of water on demand.

The project was developed as part of the Engineering Workshop Lab semester requirement, with the goal of applying embedded systems knowledge to a functional prototype. The system is intuitive — a user selects the desired volume on a menu displayed on an OLED screen, places a glass in front of the sensor, and the machine handles the rest automatically.

3. Objectives

- To design and build a functional automatic water dispensing system using Arduino UNO.
- To implement a user-friendly OLED menu interface for selecting dispensing volumes (100 ml, 150 ml, 200 ml, and No Limit).
- To utilize an Ultrasonic Sensor for automatic glass detection and trigger-based pumping.
- To implement a No Limit mode where the pump runs continuously until the glass is removed or the IR sensor detects a full glass.
- To integrate an Infrared (IR) Sensor to detect water level in the glass and prevent overflow in No Limit mode.
- To integrate a Water Level Sensor to monitor and report low water conditions.
- To provide audio and visual alerts (buzzer and LED) when the reservoir is running low.

- To demonstrate practical skills in circuit design, component integration, and embedded C programming.

4. Components Used

Component	Quantity	Purpose
Arduino UNO	1	Main microcontroller — brain of the system
OLED Display (I2C)	1	Displays volume selection menu and alerts
Ultrasonic Sensor (HC-SR04)	1	Detects glass placement in front of dispenser
Water Level Sensor	1	Monitors water level in the reservoir
Mini Submersible Water Pump	1	Pumps and dispenses water
Single Channel Relay Module	1	Controls the water pump via Arduino signal
Active Buzzer	1	Beeps when water level is critically low
LED	1	Visual indicator for low water warning
Resistor	1	Current limiting for the LED
Capacitor	1	Stabilizes power supply to prevent pump interference
Infrared (IR) Sensor	1	Detects when glass is full in No Limit mode and stops the pump

5. System Design & How It Works

5.1 Menu Selection

Upon powering the system, the OLED display activates and presents the user with a menu of four dispensing options: 100 ml, 150 ml, 200 ml, and No Limit. A physical push button is connected to the Arduino and allows the user to cycle through these options. The currently selected option is highlighted on screen, making navigation simple and intuitive.

5.2 Glass Detection & Dispensing

Once the desired volume is selected, the Ultrasonic Sensor — mounted at the front of the dispenser — continuously measures the distance in front of it. When a glass is placed within the detection range, the sensor triggers the Arduino to activate the relay module, which in turn switches on the submersible water pump. The pump runs for a precisely calculated duration corresponding to the selected volume (100 ml, 150 ml, or 200 ml), then automatically shuts off. This ensures accurate and repeatable dispensing without any user intervention beyond placing the glass.

5.3 No Limit Mode & Infrared Sensor

In addition to the three fixed-volume options, the system includes a fourth menu option: No Limit. When this mode is selected, the pump does not run for a fixed duration. Instead, it continues dispensing water for as long as the glass remains in front of the Ultrasonic Sensor. The pump stops automatically the moment the glass or hand is removed from the detection zone.

To prevent overflow in No Limit mode, an Infrared (IR) Sensor is mounted at a fixed height on the dispenser positioned at the level where the glass is considered full. As the water rises and reaches that level, the IR sensor detects the water surface and immediately signals the Arduino to cut off the pump, preventing spillage. This makes the No Limit mode both flexible for larger containers and safe against overflow.

5.3 Water Level Monitoring & Alerts

A Water Level Sensor is submerged in the water reservoir. It continuously sends readings to the Arduino. When the water level drops below a defined threshold, the system triggers a multi-modal alert: the OLED display shows a 'Water Low — Please Refill' message, the Active Buzzer emits a repeated beeping tone, and the LED lights up as a visible indicator. This ensures the user is promptly informed before the reservoir runs completely dry, preventing the pump from running without water (dry running), which can damage the pump.

6. Circuit Overview

The circuit is centered around the Arduino UNO, which serves as the central processing unit. All sensor inputs and output devices are connected to its digital and analog pins. The following summarizes the key connections:

- The OLED display communicates with the Arduino via the I2C protocol (SDA and SCL pins).

- The Ultrasonic Sensor uses two digital pins — one for trigger and one for echo — to measure distance.
- The Water Level Sensor is connected to an analog input pin to read varying voltage levels based on water depth.
- The Relay Module is connected to a digital output pin; it switches the pump's power circuit on and off.
- The Infrared (IR) Sensor is connected to a digital input pin; it detects when water reaches the full level in the glass during No Limit mode and signals the Arduino to stop the pump.
- The Active Buzzer and LED (with current-limiting resistor) are connected to separate digital output pins.
- A capacitor is placed across the pump's power supply lines to filter voltage spikes and stabilize the circuit.

Please refer to the Project Pictures folder in the GitHub repository for photographs of the physical circuit and assembled prototype.

7. Challenges & Problems Faced

During the development and testing phase of the project, the team encountered several technical challenges. These are documented below along with the steps taken to address them.

7.1 Water Pump Interference with Ultrasonic Sensor

The water pump caused erratic readings from the Ultrasonic Sensor due to voltage fluctuations and electrical noise on the power lines when the pump activated. Adding a capacitor across the pump's power terminals filtered the noise and stabilized the sensor readings, resolving the issue.

7.2 OLED Display Menu Glitching

The OLED menu intermittently glitched during operation — characters flickered, the screen froze, or the menu failed to update on button press. The team investigated I2C communication speed, refresh rate, and button debouncing, and made partial improvements through software debouncing and adjusting the display update logic. However, the issue was not fully resolved and is likely caused by the screen being refreshed too frequently in the main loop. Optimizing the code to only redraw on change remains a future improvement.

8. Industrial & Real-World Applications

While this project was developed as a prototype for academic purposes, the underlying concept has significant potential for real-world and industrial applications. Below are several domains where an automated precision dispensing system of this nature could be deployed:

8.1 Automated Fuel Dispensing

The most direct industrial parallel to this project is an automated fuel dispensing system — similar in concept to a petrol pump. Instead of water, the system could be adapted to dispense precise quantities of fuel, oil, or other liquids into vehicles or machinery. Sensors would detect the nozzle or container placement, and the user could select the desired volume or amount via a display panel. This would improve accuracy, reduce wastage, and minimize human error at fuel stations or in industrial refueling operations.

8.2 Pharmaceutical & Medical Dispensing

In hospitals and pharmaceutical manufacturing, precise liquid measurement is critical. A scaled and calibrated version of this system could be used to dispense exact doses of liquid medicines, chemical reagents, or IV fluids. The sensor-based automation would reduce contamination risks associated with manual handling and improve dosage accuracy.

8.3 Food & Beverage Industry

Automated dispensers are widely used in the food and beverage industry for portion control and consistency. This system could be adapted for dispensing beverages, syrups, sauces, or other liquid ingredients in precise amounts, improving consistency in food production lines and reducing waste.

8.4 Smart Water Management

In water-scarce regions or smart home environments, a system like this could be integrated into water management infrastructure to dispense controlled amounts of water for drinking, irrigation, or industrial cooling — helping conserve water and track consumption in real time.

9. Conclusion

The Automatic Water Dispenser project successfully demonstrates the integration of multiple electronic components under the control of an Arduino UNO microcontroller to create a functional and practical automated system. The project achieves its core objectives: precise volume selection through an OLED menu, automatic glass detection via an Ultrasonic Sensor, reliable pump control through a relay, and water level monitoring with audio-visual alerts.

The challenges encountered particularly the pump interference issue resolved with a capacitor, and the OLED glitching provided valuable hands-on learning experiences in circuit debugging, noise filtering, and embedded software optimization. These experiences reinforced the importance of understanding hardware interactions and not just software logic in embedded systems design.

With further development, this prototype has the potential to be scaled into a commercially viable product with applications in the fuel, medical, food, and water management industries. The project has been a rewarding demonstration of how embedded systems and sensor integration can be applied to solve real-world problems.

10. References & Repository

- Arduino Official Documentation — arduino.cc
- Adafruit SSD1306 OLED Library Documentation
- HC-SR04 Ultrasonic Sensor Datasheet
- Project GitHub Repository: github.com/sybaushazi/Automatic-Dispenser

11. Team Members

Name	Roll Number	Contribution
Shahzaib Sohail	25i-6649	Documentation, GitHub Setup, Code Review
Mateen Ahmed	25i-6614	Circuit Design,Component Testing, Arduino Programming